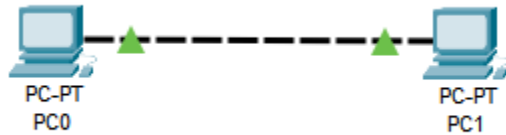


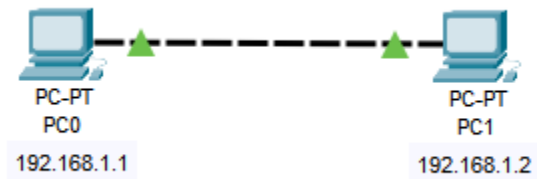
simple network

PC0 --- PC1

1. Setup the connection



2. Assign IP addresses



3. Ping PC0 to PC1

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

4. Ping PC1 to PC0

```
C:\>ping 192.168.1.1

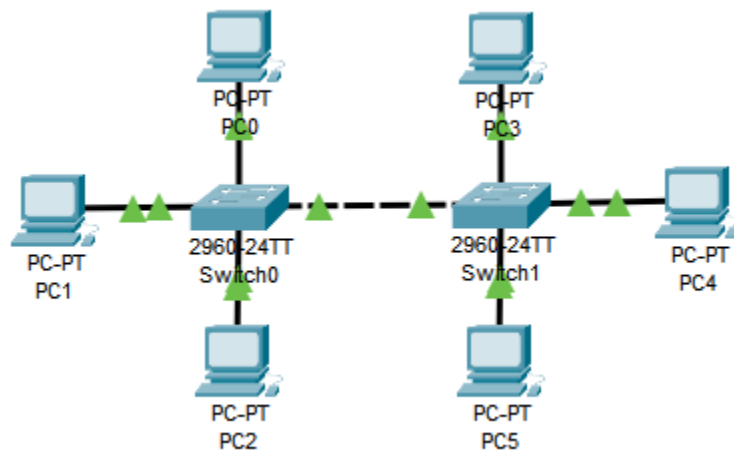
Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

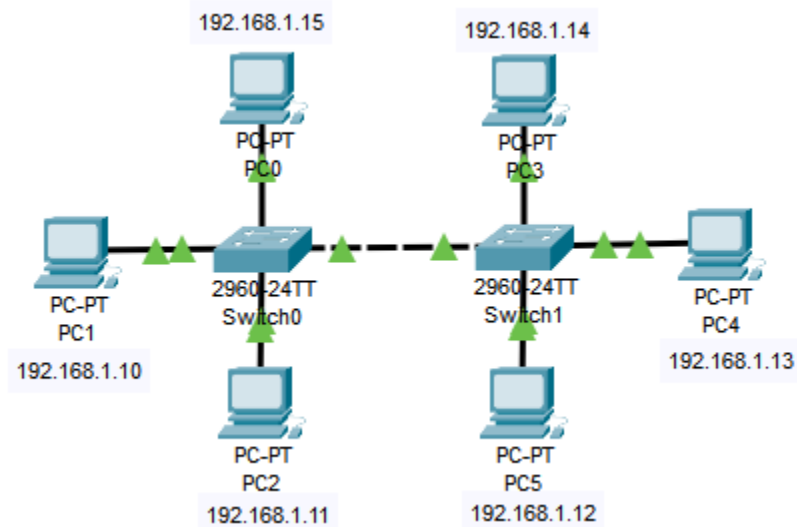
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

network switch

1. Setup the connection



2. Assign IP addresses (all PCs with the same network address but different host addresses for each PC)



3. Ping PCs
PC5 to PC0

```
C:\>ping 192.168.1.15

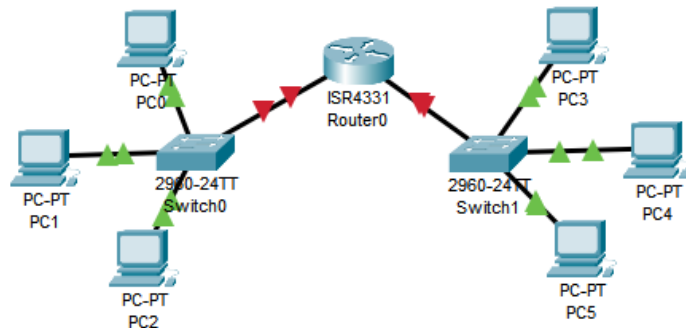
Pinging 192.168.1.15 with 32 bytes of data:

Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=3ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128

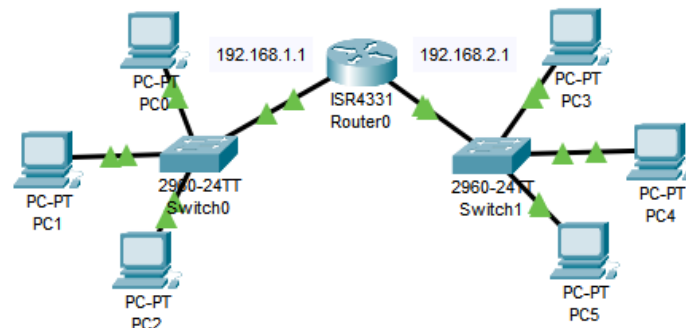
Ping statistics for 192.168.1.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

connect two different networks using router

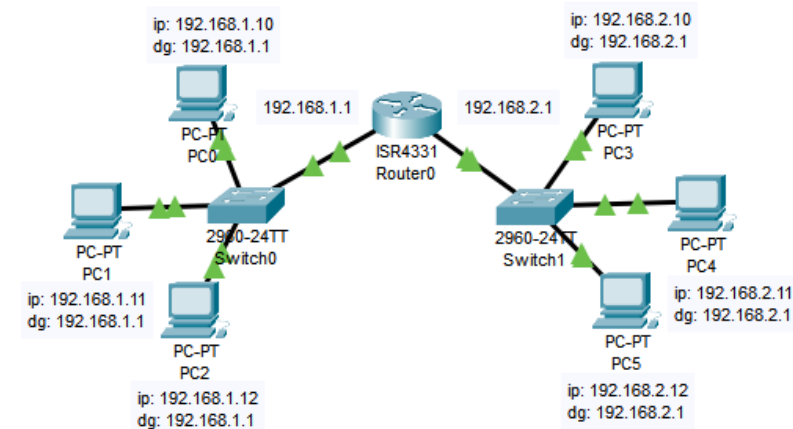
1. Setup the connection



2. Router configuration



3. Assign IP addresses and default gateways to PCs



4. Ping test

```
C:\>ping 192.168.2.12

Pinging 192.168.2.12 with 32 bytes of data:

Reply from 192.168.2.12: bytes=32 time<1ms TTL=127
Reply from 192.168.2.12: bytes=32 time=3ms TTL=127
Reply from 192.168.2.12: bytes=32 time<1ms TTL=127
Reply from 192.168.2.12: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Router Configuration

1. CLI

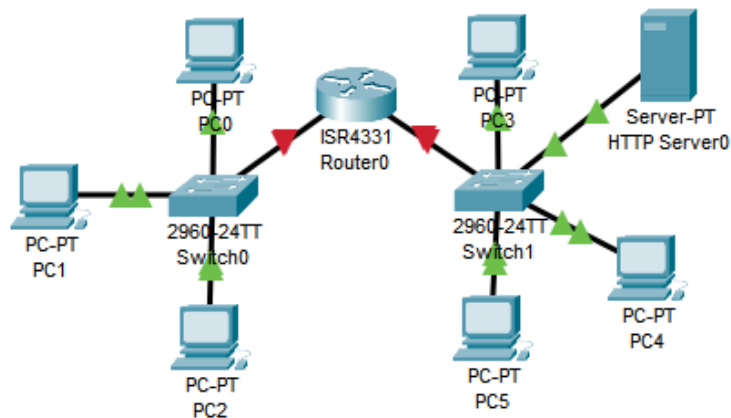
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gig0/0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

2. Config

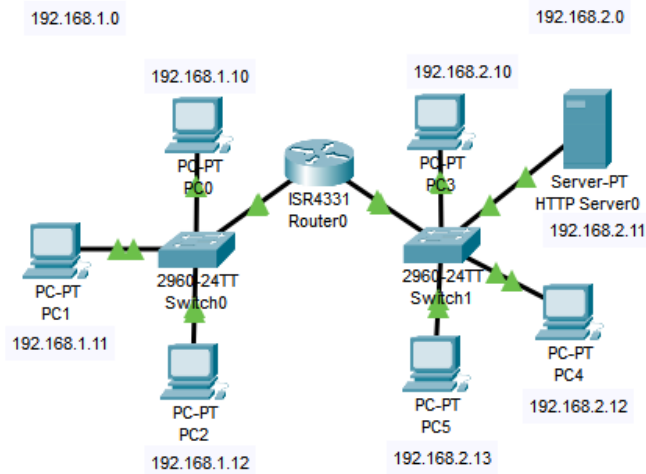
Port Status	☒ On
Bandwidth	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0001.43BE.0701
IP Configuration	
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0

HTTP without DNS server

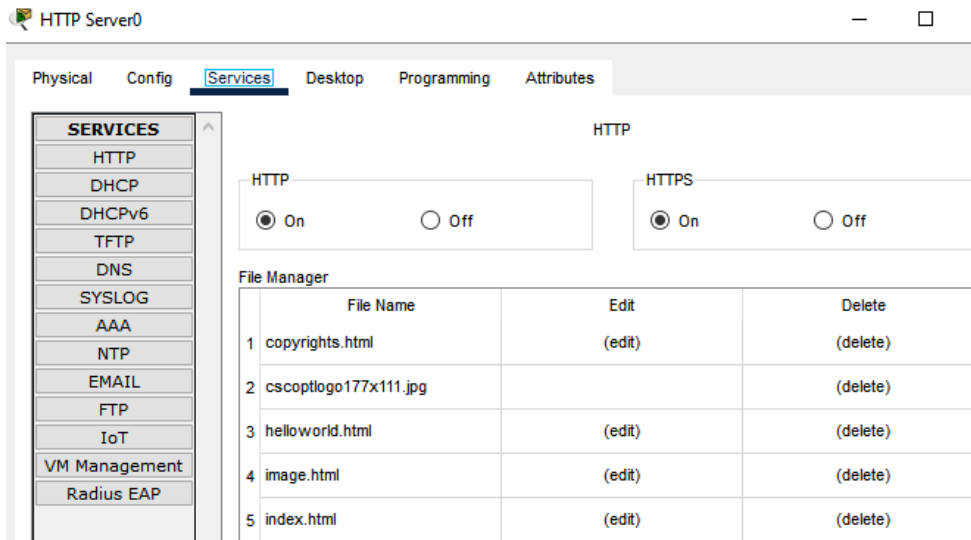
1. Setup the connection



- Assign IP addresses, router configuration, and default gateways



- Server -> Services -> HTTP -> Services ON -> edit index.html

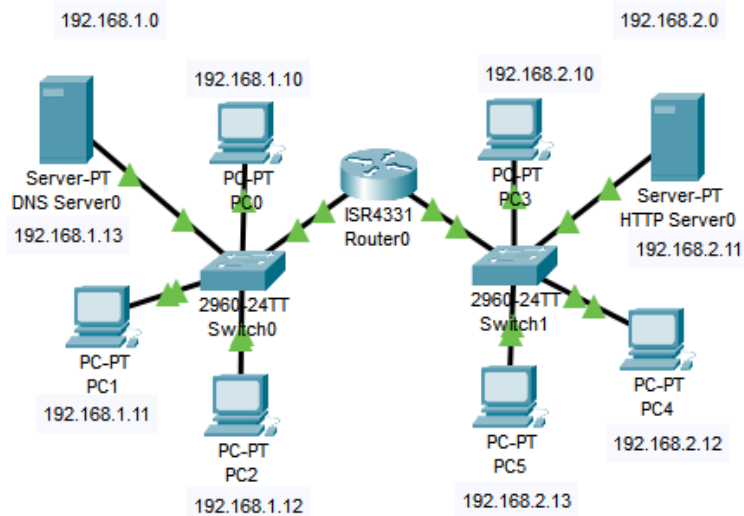


- PC -> desktop -> browser UML -> Server IP HTTP

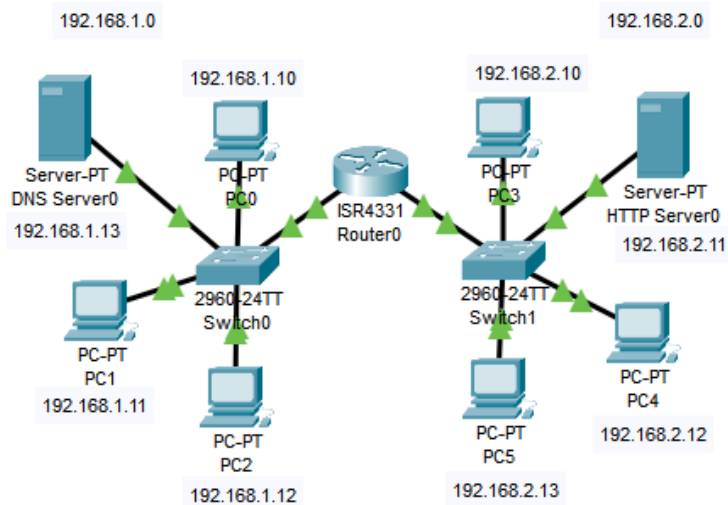


DNS Server Implementation

1. Setup the connection



2. Assign IP addresses, router configuration, and default gateways



3. DNS Server -> Services -> DNS On -> enter name and IP -> add

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type A Record

Address

No.	Name	Type	Detail
0	web1.com	A Record	192.168.2.11

- assign DNS server IP address to all devices (PCs + Servers) -> DNS server

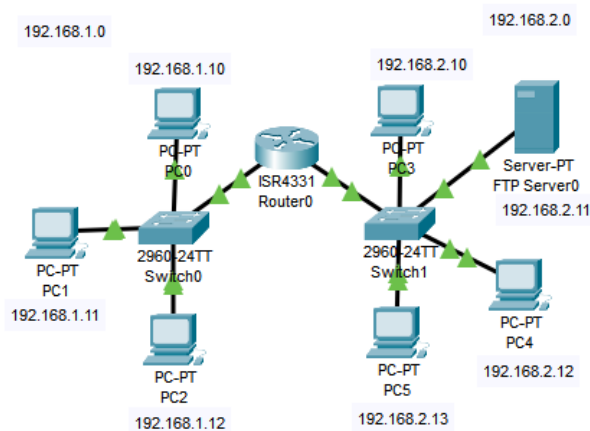
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.13

- PC0 -> Desktop -> UML browser -> enter name of website

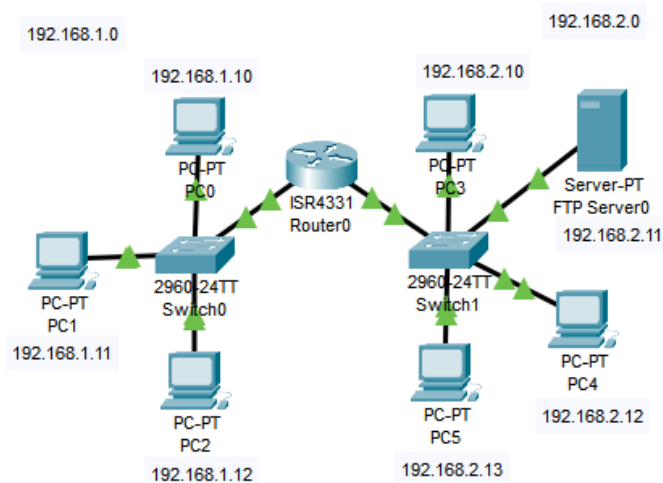


FTP (File Transfer Protocol)

- Setup the connection



- assign IP address, router configuration and default gateways



3. Server -> services -> FTP -> FTP On -> set username and password and permit rwd

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☒ Write ☒ Read ☒ Delete ☒ Rename ☒ List

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	naveen	123	RWDNL

4. PC -> desktop -> text editor -> create and save file

PC1

Physical Config **Desktop**

Text Editor

File

hello world

5. PC -> desktop -> command prompt

commands:

ftp <ftp server ip address>

put <filename>

dir

PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.2.11
Trying to connect...192.168.2.11
Connected to 192.168.2.11
220- Welcome to FT Ftp server
Username:naveen
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put naveen.txt

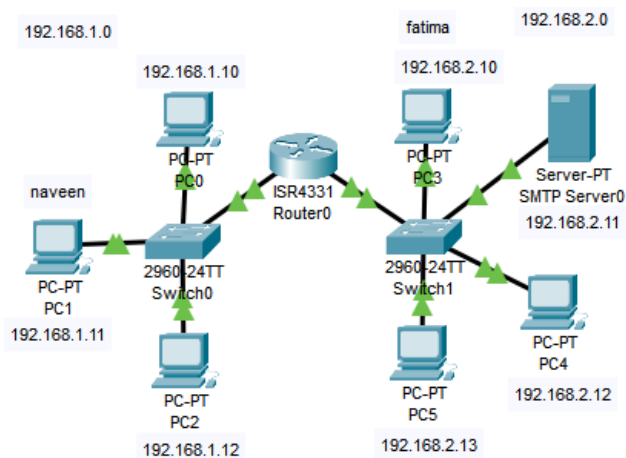
Writing file naveen.txt to 192.168.2.11:
File transfer in progress...

[Transfer complete - 11 bytes]

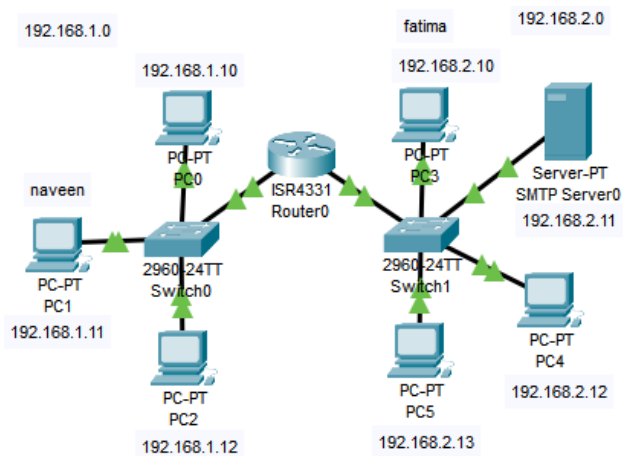
11 bytes copied in 0.089 secs (123 bytes/sec)
ftp>dir
```


SMTP (Simple Mail Transfer Protocol)

1. Setup the connection



2. assign IP addresses, router configurations, and default gateways



3. Server -> Services -> EMAIL -> turn on -> enter domain (eg. gmail.com) -> username and password (eg. naveen 123)

EMAIL

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

Domain Name:

User Setup

User Password

4. PCX -> desktop -> email -> configure mail -> enter all -> outgoing and incoming IP : server ip on which email services setup

PC1

Physical	Config	Desktop	Program
Configure Mail			
User Information			
Your Name:		naveen	
Email Address		naveen@gmail.com	
Server Information			
Incoming Mail Server		192.168.2.11	
Outgoing Mail Server		192.168.2.11	
Logon Information			
User Name:		naveen	
Password:		...	

5. PCY -> desktop -> email -> configure mail -> enter all -> outgoing and incoming IP : server ip on which email services setup

PC3

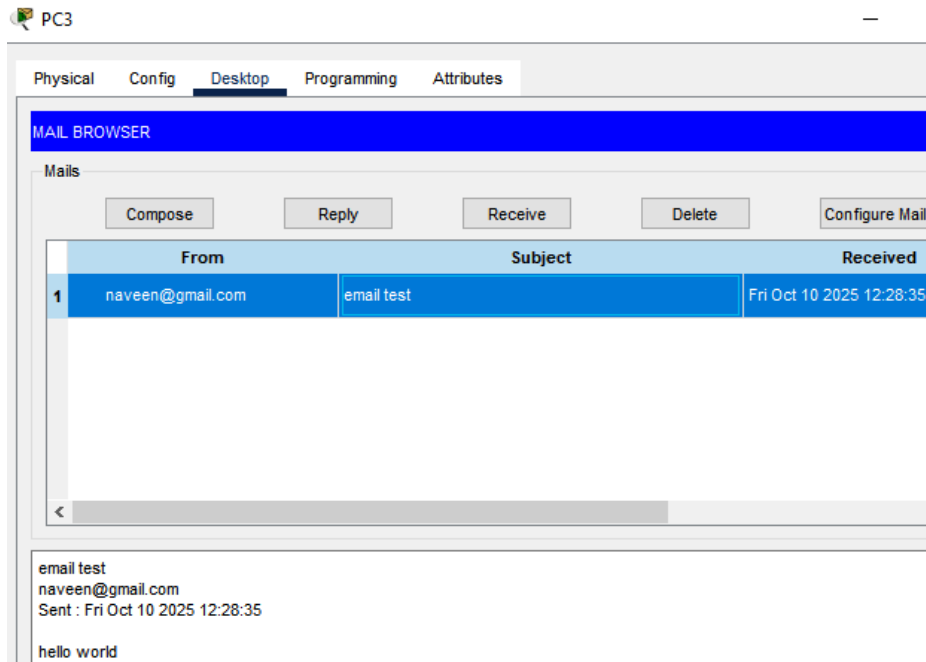
Physical	Config	Desktop	Progr
Configure Mail			
User Information			
Your Name:		fatima	
Email Address		fatima@gmail.com	
Server Information			
Incoming Mail Server		192.168.2.11	
Outgoing Mail Server		192.168.2.11	
Logon Information			
User Name:		fatima	
Password:		...	

6. PCX -> compose email -> send

PC1

Physical	Config	Desktop	Programmi
Compose Mail			
Send		To: fatima@gmail.com	
		Subject: email test	
hello world			

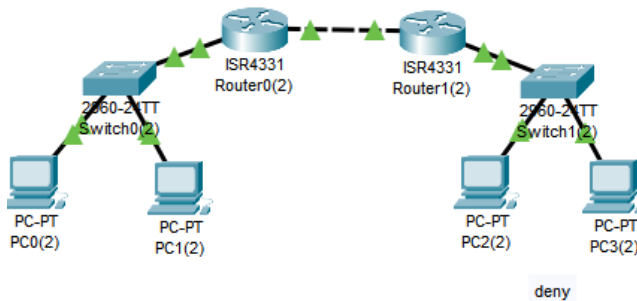
7. PCY -> email -> receive email -> received



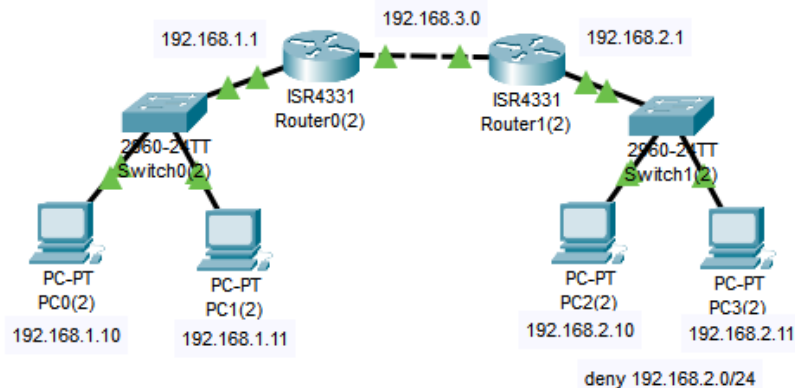
ACL (Access Control List)

Scenario : allow 192.168.1.0/24 to 192.168.2.0/24, deny 192.168.2.0/24 to 192.168.1.0/24

1. setup the connection



2. assign IP addresses, router configurations, default gateways, and ip route



3. decide standard/extended, number, interface, in/out
4. selected router -> CLI -> enter commands

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 101 permit icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
echo
Router(config)#access-list 101 permit icmp 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
echo-reply
Router(config)#access-list 101 deny icmp 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
echo
Router(config)#access-list 101 permit ip any any
Router(config)#interface gig0/0/1
Router(config-if)#ip access-group 101 in
Router(config-if)#end
Router#
%SYS-5-CONFIG I: Configured from console by console
```

commands (standard):

```
en
conf t
access-list 1 deny <source IP address> 0.0.0.0
access-list 1 permit any
interface <selected interface> -> nearer to the destination
ip access-group 1 out
```

commands (extended):

```
en
conf t
access-list 120 <source IP> 0.0.0.0 <destination IP> 0.0.0.255 (if dest. whole network)
access-list 120 permit any any
interface <selected interface> -> nearer to the source
ip access-group 120 in
```

commands (ICMP standard/extended)

scenario 1:

allow A -> B, deny B -> A

Network A: 192.168.1.0/24

Network B: 192.168.2.0/24

```
access-list 101 permit icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 echo
access-list 101 permit icmp 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255 echo-reply
access-list 101 deny icmp 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255 echo
access-list 101 permit ip any any
```

```
interface gig0/0/1
ip access-group 101 in
end
```

scenario 2:

allow B -> A, deny A -> B

Network A: 192.168.1.0/24

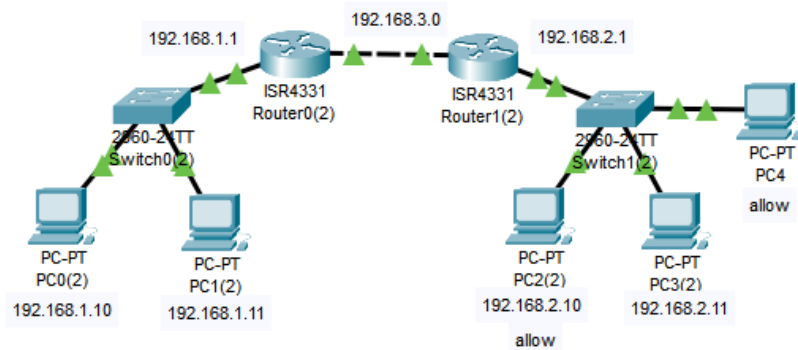
Network B: 192.168.2.0/24

```
access-list 120 permit icmp 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255 echo
access-list 120 permit icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 echo-reply
access-list 120 deny icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 echo
access-list 120 permit any any
```

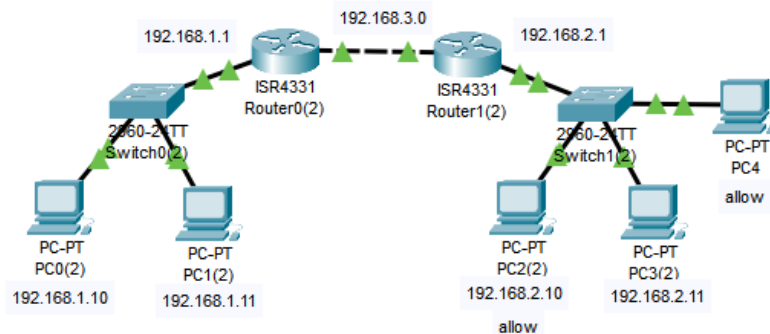
```
interface <interface>
ip access-group 120 in
end
```

Scenario : deny 1 PC, and allow 2 PCs from 192.168.2.0/24 to 192.168.1.0/24

1. Setup the connection



2. Assign IP addresses, router configurations, default gateways, and ip route



3. Decide standard/extended, number, in/out
4. Selected router -> CLI -> enter following commands

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 120 permit icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
echo
Router(config)#access-list 120 permit icmp 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
echo-reply
Router(config)#access-list 120 deny 192.168.2.11 0.0.0.0 192.168.1.0 0.0.0.255 echo
^
% Invalid input detected at '^' marker.

Router(config)#access-list 120 deny icmp 192.168.2.11 0.0.0.0 192.168.1.0 0.0.0.255 echo
Router(config)#access-list 120 permit any any
^
% Invalid input detected at '^' marker.

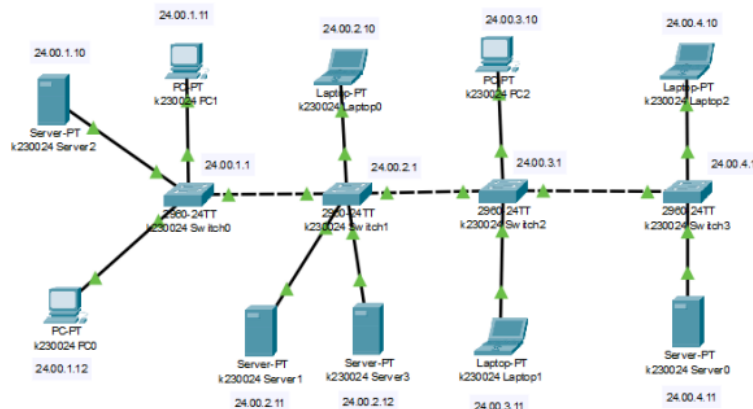
Router(config)#access-list 120 permit any
^
% Invalid input detected at '^' marker.

Router(config)#access-list 120 permit ip any
% Incomplete command.
Router(config)#access-list 120 permit ip any any
Router(config)#interface gig0/0/1
Router(config-if)#ip access-group 120 in
Router(config-if)#end

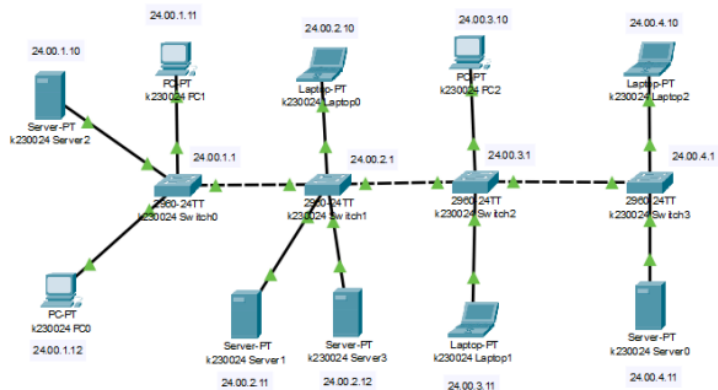
```

Telnet

1. Setup the connection



2. Assign IP addresses, gateways, router configurations



3. switch/router -> CLI -> telnet configuration commands

k230024 Switch0

```
Physical  Config  CLI  Attributes
IOS Command Line Interface

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname k230024
k230024(config)#enable secret pass
k230024(config)#interface vlan 1
k230024(config-if)#ip add 24.00.1.1 255.0.0.0
k230024(config-if)#no shutdown

k230024(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

k230024(config-if)#exit
k230024(config)#line vty 0 4
k230024(config-line)#password pass1
k230024(config-line)#login
k230024(config-line)#exit
k230024(config)#
```

4. PC -> command prompt -> telnet 'switch IP'

k230024 PC1

```
Physical  Config  Desktop  Programming  Attr
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 24.00.1.1
Trying 24.0.1.1 ...Open

User Access Verification

Password:
k230024>en
Password:
k230024#
```

5. Enter password and login

k230024 PC1

```
Physical  Config  Desktop  Programming  Attr
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 24.00.1.1
Trying 24.0.1.1 ...Open

User Access Verification

Password:
k230024>en
Password:
k230024#
```

6. en -> can now config that switch/router through a PC

SSH

1. Setup the connection
2. Assign IP addresses, gateways, router configurations
3. switch/router -> CLI -> ssh configuration commands
4. PC -> command prompt -> ssh -l 'username' 'switch ip'
5. enter password & login